Chapter 2

Minimum Variance Unbiased Estimation

2.1 Introduction

In this chapter we will begin our search for good estimators of unknown deterministic parameters. We will restrict our attention to estimators which on the average yield the true parameter value. Then, within this class of estimators the goal will be to find the one that exhibits the least variability. Hopefully, the estimator thus obtained will produce values close to the true value most of the time. The notion of a minimum variance unbiased estimator is examined within this chapter, but the means to find it will require some more theory. Succeeding chapters will provide that theory as well as apply it to many of the typical problems encountered in signal processing.

2.2 Summary

An unbiased estimator is defined by (2.1), with the important proviso that this holds for all possible values of the unknown parameter. Within this class of estimators the one with the minimum variance is sought. The unbiased constraint is shown by example to be desirable from a practical viewpoint since the more natural error criterion, the *minimum mean square error*, defined in (2.5), generally leads to unrealizable estimators. Minimum variance unbiased estimators do not, in general, exist. When they do, several methods can be used to find them. The methods rely on the Cramer-Rao lower bound and the concept of a sufficient statistic. If a minimum variance unbiased estimator does not exist or if both of the previous two approaches fail, a further constraint on the estimator, to being linear in the data, leads to an easily implemented, but suboptimal, estimator.

2.3 Unbiased Estimators

For an estimator to be unbiased we mean that *on the average* the estimator will yield the true value of the unknown parameter. Since the parameter value may in general be anywhere in the interval $a < \theta < b$, unbiasedness asserts that no matter what the true value of θ, our estimator will yield it *on the average.* Mathematically, an estimator is *unbiased* if

$$E(\hat{\theta}) = \theta \qquad a < \theta < b \tag{2.1}$$

where (a, b) denotes the range of possible values of θ.

Example 2.1 - Unbiased Estimator for DC Level in White Gaussian Noise

Consider the observations

$$x[n] = A + w[n] \qquad n = 0, 1, \ldots, N-1$$

where A is the parameter to be estimated and $w[n]$ is WGN. The parameter A can take on any value in the interval $-\infty < A < \infty$. Then, a reasonable estimator for the average value of $x[n]$ is

$$\hat{A} = \frac{1}{N} \sum_{n=0}^{N-1} x[n] \tag{2.2}$$

or the sample mean. Due to the linearity properties of the expectation operator

$$\begin{aligned} E(\hat{A}) &= E\left[\frac{1}{N} \sum_{n=0}^{N-1} x[n]\right] \\ &= \frac{1}{N} \sum_{n=0}^{N-1} E(x[n]) \\ &= \frac{1}{N} \sum_{n=0}^{N-1} A \\ &= A \end{aligned}$$

for all A. The sample mean estimator is therefore unbiased. ◇

In this example A can take on any value, although in general the values of an unknown parameter may be restricted by physical considerations. Estimating the resistance R of an unknown resistor, for example, would necessitate an interval $0 < R < \infty$.

Unbiased estimators tend to have symmetric PDFs centered about the true value of θ, although this is not necessary (see Problem 2.5). For Example 2.1 the PDF is shown in Figure 2.1 and is easily shown to be $\mathcal{N}(A, \sigma^2/N)$ (see Problem 2.3).

The restriction that $E(\hat{\theta}) = \theta$ *for all* θ is an important one. Letting $\hat{\theta} = g(\mathbf{x})$, where $\mathbf{x} = [x[0]\, x[1] \ldots x[N-1]]^T$, it asserts that

$$E(\hat{\theta}) = \int g(\mathbf{x}) p(\mathbf{x}; \theta)\, d\mathbf{x} = \theta \qquad \text{for all } \theta. \tag{2.3}$$

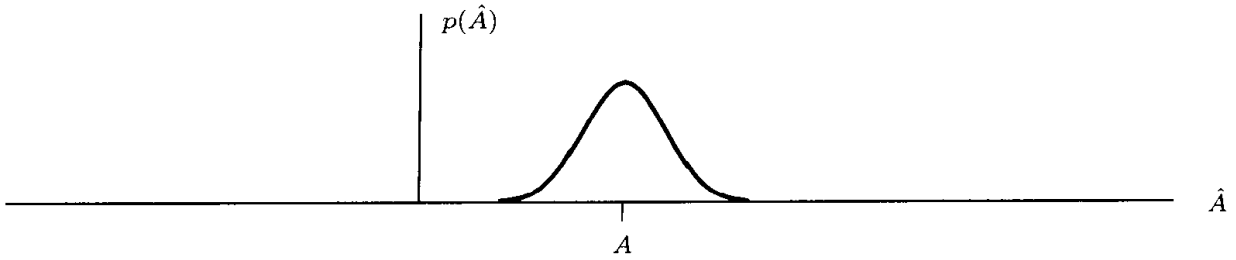


Figure 2.1 Probability density function for sample mean estimator

It is possible, however, that (2.3) may hold for some values of θ and not others, as the next example illustrates.

Example 2.2 - Biased Estimator for DC Level in White Noise

Consider again Example 2.1 but with the *modified* sample mean estimator

$$\check{A} = \frac{1}{2N} \sum_{n=0}^{N-1} x[n].$$

Then,

$$\begin{aligned} E(\check{A}) &= \frac{1}{2}A \\ &= A \text{ if } A = 0 \\ &\neq A \text{ if } A \neq 0. \end{aligned}$$

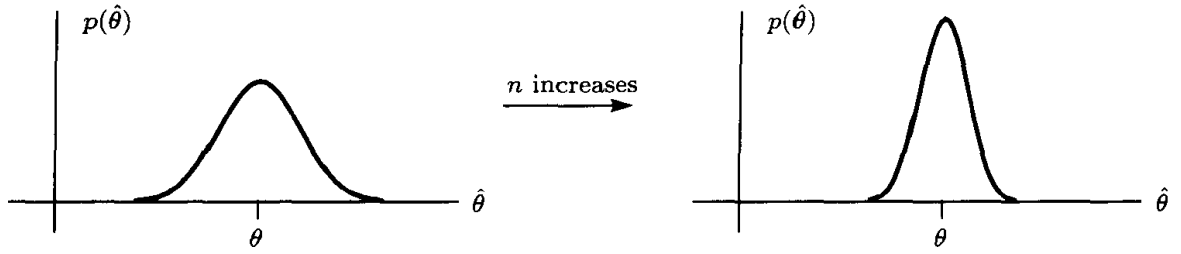
It is seen that (2.3) holds for the modified estimator only for $A = 0$. Clearly, $\check{A}$ is a biased estimator. ◇

That an estimator is unbiased does not necessarily mean that it is a good estimator. It only guarantees that on the average it will attain the true value. On the other hand, biased estimators are ones that are characterized by a systematic error, which presumably should not be present. A persistent bias will always result in a poor estimator. As an example, the unbiased property has an important implication when several estimators are combined (see Problem 2.4). It sometimes occurs that multiple estimates of the same parameter are available, i.e., $\{\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_n\}$. A reasonable procedure is to combine these estimates into, hopefully, a better one by averaging them to form

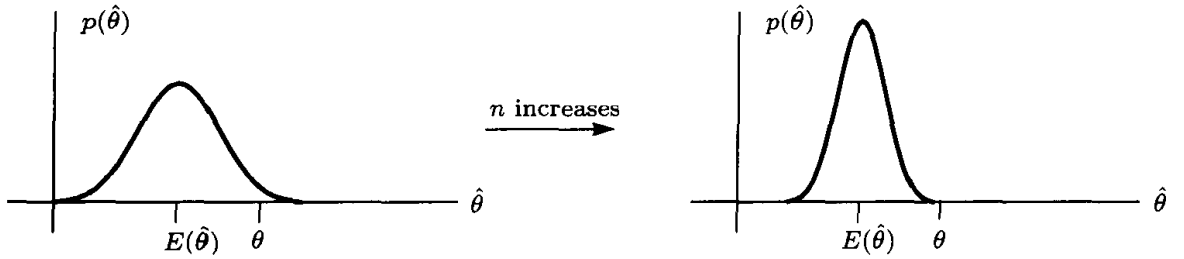
$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^{n} \hat{\theta}_i. \tag{2.4}$$

Assuming the estimators are unbiased, with the same variance, and uncorrelated with each other,

$$E(\hat{\theta}) = \theta$$



(a) Unbiased estimator



(b) Biased estimator

Figure 2.2 Effect of combining estimators

and

$$\begin{aligned} \text{var}(\hat{\theta}) &= \frac{1}{n^2}\sum_{i=1}^{n}\text{var}(\hat{\theta}_i) \\ &= \frac{\text{var}(\hat{\theta}_1)}{n} \end{aligned}$$

so that as more estimates are averaged, the variance will decrease. Ultimately, as $n \to \infty$, $\hat{\theta} \to \theta$. However, if the estimators are biased or $E(\hat{\theta}_i) = \theta + b(\theta)$, then

$$\begin{aligned} E(\hat{\theta}) &= \frac{1}{n}\sum_{i=1}^{n}E(\hat{\theta}_i) \\ &= \theta + b(\theta) \end{aligned}$$

and no matter how many estimators are averaged, $\hat{\theta}$ will not converge to the true value. This is depicted in Figure 2.2. Note that, in general,

$$b(\theta) = E(\hat{\theta}) - \theta$$

is defined as the *bias* of the estimator.

2.4 Minimum Variance Criterion

In searching for optimal estimators we need to adopt some optimality criterion. A natural one is the *mean square error* (MSE), defined as

$$\text{mse}(\hat{\theta}) = E\left[(\hat{\theta} - \theta)^2\right]. \tag{2.5}$$

This measures the average mean squared deviation of the estimator from the true value. Unfortunately, adoption of this natural criterion leads to unrealizable estimators, ones that cannot be written solely as a function of the data. To understand the problem which arises we first rewrite the MSE as

$$\begin{aligned} \text{mse}(\hat{\theta}) &= E\left\{\left[\left(\hat{\theta} - E(\hat{\theta})\right) + \left(E(\hat{\theta}) - \theta\right)\right]^2\right\} \\ &= \text{var}(\hat{\theta}) + \left[E(\hat{\theta}) - \theta\right]^2 \\ &= \text{var}(\hat{\theta}) + b^2(\theta) \end{aligned} \tag{2.6}$$

which shows that the MSE is composed of errors due to the variance of the estimator as well as the bias. As an example, for the problem in Example 2.1 consider the modified estimator

$$\check{A} = a\frac{1}{N}\sum_{n=0}^{N-1} x[n]$$

for some constant a. We will attempt to find the a which results in the minimum MSE. Since $E(\check{A}) = aA$ and $\text{var}(\check{A}) = a^2\sigma^2/N$, we have, from (2.6),

$$\text{mse}(\check{A}) = \frac{a^2\sigma^2}{N} + (a-1)^2A^2.$$

Differentiating the MSE with respect to a yields

$$\frac{d\,\text{mse}(\check{A})}{da} = \frac{2a\sigma^2}{N} + 2(a-1)A^2$$

which upon setting to zero and solving yields the optimum value

$$a_{\text{opt}} = \frac{A^2}{A^2 + \sigma^2/N}.$$

It is seen that, unfortunately, the optimal value of a depends upon the unknown parameter A. The estimator is therefore not realizable. In retrospect the estimator depends upon A since the bias term in (2.6) is a function of A. It would seem that any criterion which depends on the bias will lead to an unrealizable estimator. Although this is generally true, on occasion realizable minimum MSE estimators can be found [Bibby and Toutenburg 1977, Rao 1973, Stoica and Moses 1990].

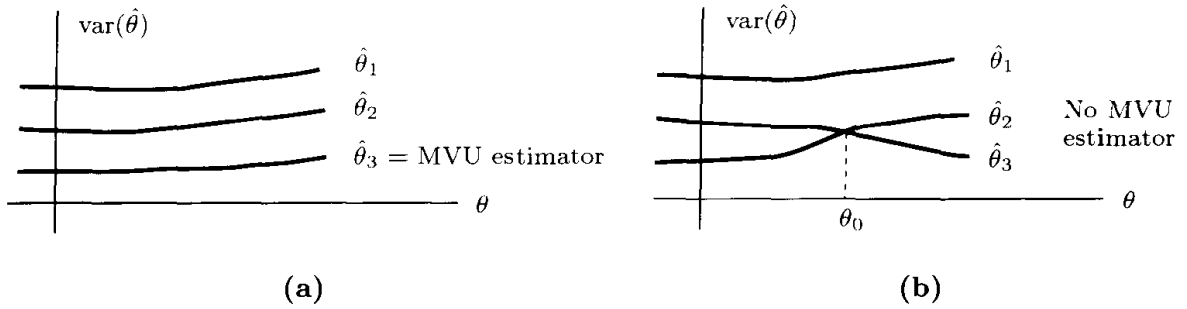


Figure 2.3 Possible dependence of estimator variance with θ

From a practical viewpoint the minimum MSE estimator needs to be abandoned. An alternative approach is to constrain the bias to be zero and find the estimator which minimizes the variance. Such an estimator is termed the *minimum variance unbiased* (MVU) estimator. Note that from (2.6) that the MSE of an unbiased estimator is just the variance.

Minimizing the variance of an unbiased estimator also has the effect of concentrating the PDF of the estimation error, $\hat{\theta} - \theta$, about zero (see Problem 2.7). The estimation error will therefore be less likely to be large.

2.5 Existence of the Minimum Variance Unbiased Estimator

The question arises as to whether a MVU estimator exists, i.e., an unbiased estimator with minimum variance for all θ. Two possible situations are described in Figure 2.3. If there are three unbiased estimators that exist and whose variances are shown in Figure 2.3a, then clearly $\hat{\theta}_3$ is the MVU estimator. If the situation in Figure 2.3b exists, however, then there is no MVU estimator since for $\theta < \theta_0$, $\hat{\theta}_2$ is better, while for $\hat{\theta} > \theta_0$, $\hat{\theta}_3$ is better. In the former case $\hat{\theta}_3$ is sometimes referred to as the *uniformly minimum variance unbiased* estimator to emphasize that the variance is smallest *for all* θ. In general, the MVU estimator *does not always exist*, as the following example illustrates.

Example 2.3 - Counterexample to Existence of MVU Estimator

If the *form* of the PDF changes with θ, then it would be expected that the best estimator would also change with θ. Assume that we have two independent observations $x[0]$ and $x[1]$ with PDF

$$\begin{aligned} x[0] &\sim \mathcal{N}(\theta, 1) \\ x[1] &\sim \begin{cases} \mathcal{N}(\theta, 1) & \text{if } \theta \geq 0 \\ \mathcal{N}(\theta, 2) & \text{if } \theta < 0. \end{cases} \end{aligned}$$

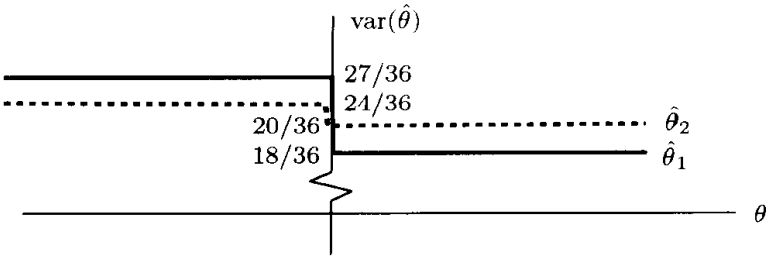


Figure 2.4 Illustration of nonexistence of minimum variance unbiased estimator

The two estimators

$$\begin{aligned}\hat{\theta}_1 &= \frac{1}{2}(x[0]+x[1]) \\ \hat{\theta}_2 &= \frac{2}{3}x[0]+\frac{1}{3}x[1]\end{aligned}$$

can easily be shown to be unbiased. To compute the variances we have that

$$\begin{aligned}\text{var}(\hat{\theta}_1) &= \frac{1}{4}(\text{var}(x[0])+\text{var}(x[1])) \\ \text{var}(\hat{\theta}_2) &= \frac{4}{9}\text{var}(x[0])+\frac{1}{9}\text{var}(x[1])\end{aligned}$$

so that

$$\text{var}(\hat{\theta}_1) = \begin{cases} \frac{18}{36} & \text{if } \theta \geq 0 \\ \frac{27}{36} & \text{if } \theta < 0 \end{cases}$$

and

$$\text{var}(\hat{\theta}_2) = \begin{cases} \frac{20}{36} & \text{if } \theta \geq 0 \\ \frac{24}{36} & \text{if } \theta < 0. \end{cases}$$

The variances are shown in Figure 2.4. Clearly, between these two estimators no MVU estimator exists. It is shown in Problem 3.6 that for $\theta \geq 0$ the minimum possible variance of an unbiased estimator is 18/36, while that for $\theta < 0$ is 24/36. Hence, no single estimator can have a variance uniformly less than or equal to the minima shown in Figure 2.4. ◇

To conclude our discussion of existence we should note that it is also possible that there may not exist even a *single* unbiased estimator (see Problem 2.11). In this case any search for a MVU estimator is fruitless.

2.6 Finding the Minimum Variance Unbiased Estimator

Even if a MVU estimator exists, we may not be able to find it. There is no known "turn-the-crank" procedure which will always produce the estimator. In the next few chapters we shall discuss several possible approaches. They are:

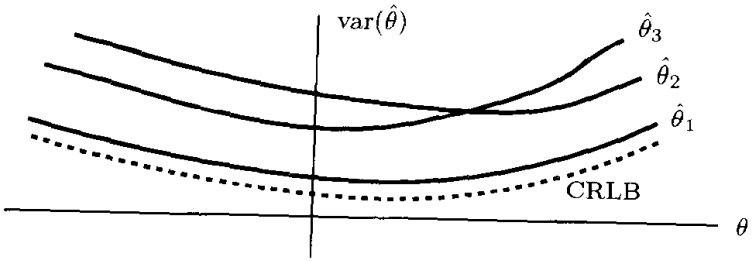


Figure 2.5 Cramer-Rao lower bound on variance of unbiased estimator

1. Determine the Cramer-Rao lower bound (CRLB) and check to see if some estimator satisfies it (Chapters 3 and 4).

2. Apply the Rao-Blackwell-Lehmann-Scheffe (RBLS) theorem (Chapter 5).

3. Further restrict the class of estimators to be not only unbiased but also *linear*. Then, find the minimum variance estimator within this restricted class (Chapter 6).

Approaches 1 and 2 may produce the MVU estimator, while 3 will yield it only if the MVU estimator is linear in the data.

The CRLB allows us to determine that for *any unbiased* estimator the variance must be greater than or equal to a given value, as shown in Figure 2.5. If an estimator exists whose variance equals the CRLB *for each value of* θ, then it must be the MVU estimator. In this case, the theory of the CRLB immediately yields the estimator. It may happen that no estimator exists whose variance equals the bound. Yet, a MVU estimator may still exist, as for instance in the case of $\hat{\theta}_1$ in Figure 2.5. Then, we must resort to the Rao-Blackwell-Lehmann-Scheffe theorem. This procedure first finds a *sufficient statistic*, one which uses all the data efficiently, and then finds a function of the sufficient statistic which is an unbiased estimator of θ. With a slight restriction of the PDF of the data this procedure will then be guaranteed to produce the MVU estimator. The third approach requires the estimator to be linear, a sometimes severe restriction, and chooses the best *linear* estimator. Of course, only for particular data sets can this approach produce the MVU estimator.

2.7 Extension to a Vector Parameter

If $\boldsymbol{\theta} = [\theta_1 \, \theta_2 \ldots \theta_p]^T$ is a vector of unknown parameters, then we say that an estimator $\hat{\boldsymbol{\theta}} = [\hat{\theta}_1 \, \hat{\theta}_2 \ldots \hat{\theta}_p]^T$ is unbiased if

$$E(\hat{\theta}_i) = \theta_i \qquad a_i < \theta_i < b_i \tag{2.7}$$

for $i = 1, 2, \ldots, p$. By defining

$$E(\hat{\boldsymbol{\theta}}) = \begin{bmatrix} E(\hat{\theta}_1) \\ E(\hat{\theta}_2) \\ \vdots \\ E(\hat{\theta}_p) \end{bmatrix}$$

we can equivalently define an unbiased estimator to have the property

$$E(\hat{\boldsymbol{\theta}}) = \boldsymbol{\theta}$$

for every $\boldsymbol{\theta}$ contained within the space defined in (2.7). A MVU estimator has the additional property that $\text{var}(\hat{\theta}_i)$ for $i = 1, 2, \ldots, p$ is minimum among all unbiased estimators.

References

Bibby, J., H. Toutenburg, *Prediction and Improved Estimation in Linear Models*, J. Wiley, New York, 1977.

Rao, C.R., *Linear Statistical Inference and Its Applications*, J. Wiley, New York, 1973.

Stoica, P., R. Moses, "On Biased Estimators and the Unbiased Cramer-Rao Lower Bound," *Signal Process.*, Vol. 21, pp. 349–350, 1990.

Problems

2.1 The data $\{x[0], x[1], \ldots, x[N-1]\}$ are observed where the $x[n]$'s are independent and identically distributed (IID) as $\mathcal{N}(0, \sigma^2)$. We wish to estimate the variance σ^2 as

$$\hat{\sigma^2} = \frac{1}{N} \sum_{n=0}^{N-1} x^2[n].$$

Is this an unbiased estimator? Find the variance of $\hat{\sigma^2}$ and examine what happens as $N \to \infty$.

2.2 Consider the data $\{x[0], x[1], \ldots, x[N-1]\}$, where each sample is distributed as $\mathcal{U}[0, \theta]$ and the samples are IID. Can you find an unbiased estimator for θ? The range of θ is $0 < \theta < \infty$.

2.3 Prove that the PDF of $\hat{A}$ given in Example 2.1 is $\mathcal{N}(A, \sigma^2/N)$.

2.4 The heart rate h of a patient is automatically recorded by a computer every 100 ms. In 1 s the measurements $\{\hat{h}_1, \hat{h}_2, \ldots, \hat{h}_{10}\}$ are averaged to obtain $\hat{h}$. If $E(\hat{h}_i) = \alpha h$ for some constant α and $\text{var}(\hat{h}_i) = 1$ for each i, determine whether averaging improves the estimator if $\alpha = 1$ and $\alpha = 1/2$. Assume each measurement is uncorrelated.

2.5 Two samples $\{x[0], x[1]\}$ are independently observed from a $\mathcal{N}(0, \sigma^2)$ distribution. The estimator

$$\hat{\sigma^2} = \frac{1}{2}(x^2[0] + x^2[1])$$

is unbiased. Find the PDF of $\hat{\sigma^2}$ to determine if it is symmetric about σ^2.

2.6 For the problem described in Example 2.1 the more general estimator

$$\hat{A} = \sum_{n=0}^{N-1} a_n x[n]$$

is proposed. Find the a_n's so that the estimator is unbiased and the variance is minimized. Hint: Use Lagrangian multipliers with unbiasedness as the constraint equation.

2.7 Two unbiased estimators are proposed whose variances satisfy $\text{var}(\hat{\theta}) < \text{var}(\check{\theta})$. If both estimators are Gaussian, prove that

$$\Pr\left\{|\hat{\theta} - \theta| > \epsilon\right\} < \Pr\left\{|\check{\theta} - \theta| > \epsilon\right\}$$

for any $\epsilon > 0$. This says that the estimator with less variance is to be preferred since its PDF is more concentrated about the true value.

2.8 For the problem described in Example 2.1 show that as $N \to \infty$, $\hat{A} \to A$ by using the results of Problem 2.3. To do so prove that

$$\lim_{N \to \infty} \Pr\left\{|\hat{A} - A| > \epsilon\right\} = 0$$

for any $\epsilon > 0$. In this case the estimator $\hat{A}$ is said to be *consistent.* Investigate what happens if the alternative estimator $\check{A} = \frac{1}{2N} \sum_{n=0}^{N-1} x[n]$ is used instead.

2.9 This problem illustrates what happens to an unbiased estimator when it undergoes a nonlinear transformation. In Example 2.1, if we choose to estimate the unknown parameter $\theta = A^2$ by

$$\hat{\theta} = \left(\frac{1}{N} \sum_{n=0}^{N-1} x[n]\right)^2,$$

can we say that the estimator is unbiased? What happens as $N \to \infty$?

2.10 In Example 2.1 assume now that in addition to A, the value of σ^2 is also unknown. We wish to estimate the vector parameter

$$\boldsymbol{\theta} = \begin{bmatrix} A \\ \sigma^2 \end{bmatrix}.$$

Is the estimator

$$\hat{\boldsymbol{\theta}} = \begin{bmatrix} \hat{A} \\ \hat{\sigma^2} \end{bmatrix} = \begin{bmatrix} \frac{1}{N} \sum_{n=0}^{N-1} x[n] \\ \frac{1}{N-1} \sum_{n=0}^{N-1} \left(x[n] - \hat{A}\right)^2 \end{bmatrix}$$

unbiased?

2.11 Given a single observation $x[0]$ from the distribution $\mathcal{U}[0, 1/\theta]$, it is desired to estimate θ. It is assumed that $\theta > 0$. Show that for an estimator $\hat{\theta} = g(x[0])$ to be unbiased we must have

$$\int_0^{\frac{1}{\theta}} g(u)\,du = 1.$$

Next prove that a function g cannot be found to satisfy this condition for all $\theta > 0$.